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Predictive athlete performance modeling with machine learning and biometric data integration

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The Purpose of this study is to propose a new integrative framework for athletic performance prediction based on state-of-the-art machine learning analysis and biometric data biometric scanning. By merging physiological signals i.e., Heart rate variability, oxygen consumption, muscle activation patterns, with psychological signals i.e., mental toughness, athlete engagement, group cohesion along with contextual training data, we create a hybrid model that performs superiorly as compared to traditional unidimensional models. Our exercisers were trained using a gradient boosting and neural network to learn the very complex non-linear relationships that exist between the physical and the psychological performance drivers. With a rich sample set of 480 athletes from different sports, the proposed model achieved 90% accuracy (R^2 = 0.90) in predicting performance outcomes and outdid the conventional methods with statistical approaches R^2 = 0.77) and machine learning based methods (R^2 = 0.77) And machine whiz methods. The robust results achieved from the model (over 90%) as compared to conventional nor statistical methods. From the analysis of the features' importance, the strongest predictor of performance are the Provided Dedicated Athletes' scores of the Functional Movement Screening (13.7%), athlete dedication (11.5%), maximum acceleration capabilities (10.2%), which verify the relationship along biomechanical, preconceived explosive power and psychological commitment. This reflects the finding whereby deep categorized athletic talent prediction requires a multi-dimensional approach by sophisticated fusion techniques. The framework is useful to coaches and sports scientists because it allows for the individualized design of injury risk mitigation and physiologically and psychologically-focused interpersonal help. This approach integrates multiple factors and constitutes an important progress in sports analytics by providing a comprehensive perspective on the complex realities which influence elite athletic performance.

Keywords Predictive athlete performance modeling, Machine learning, Biometric data integration, Sports analytics

In conjunction with the development of sensor devices, data science, and artificial intelligence, our conception of performance has evolved significantly[1]. During the previous decade, the sports industry has undergone a change in the measurement, analysis, and optimization of performance. This change has been caused by advancements in wearable devices, biometric sensors, and complex algorithms that analyze Real time multivariate data. Performance evaluation was largely considered a subjective task, but has now transformed into a high-level science of predictive analytics along with integrated physiology, biomechanics, and psychology[2].

The first efforts of predictive modeling in sports focused on isolated portions of performance, like the evaluating the probability of winning a match using statistics[1], or creating sport-specific algorithms for tactical decision making[2]. Except these early attempts did not incorporate the full scope of athletic achievement. The last few years have seen the emergence of more powerful machine learning techniques. These techniques enable the development of systems capable of simultaneous processing of multiple incoming data streams[3–5]. Advanced

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biometric monitoring systems, including Functional Movement Screen metrics for runners[6] and sensor fusion in team sports[7], have improved our ability to measure and assess physical performance to a whole new level.

While science has improved the general understanding of predictive machines, there remains a primary problem: a lack of psychological considerations in the models. Many studies in psychology, such as those on mental resilience[8] and mental toughness[9], along with athlete engagement[10,11], demonstrate their considerable role in winning competitions. Different Relational systems of these psychological variables, such as that found in the relationship between team cohesion, mental toughness, and collective attitudes among Chinese athletes[12], reveal the intricate psychosocial system within which an athlete operates. Yet, they most often remain unaccounted for in most predictions.

An athlete's psychological condition and the interpersonal relationships within the contest must be incorporated into even the most advanced biomechanical and neural network organizational schemes in order for them to function at all[4,13]. The lack of this understanding is especially clear in athletes who, despite the appropriate exercise program, experience exhaustion or a plateau in results[14,15]. It is not possible to create a reasonable model which ignores both physical and psychological components, particularly in teams when there is so much influence from coaching, group dynamics and collective identity on individual results[16,17].

Research in people analytics is evolving and offers a rung on the ladder for a potential integrative solution. Consider, for example, Mahmoud et al.[18], who developed frameworks that combine conventional performance metrics with psychological profiling to aid coaching and talent management decisions. In the same way, Taber et al.[2] created multilevel models that assess as a system all individual, team, and organizational factors at the same time. These new models illustrate a deepening understanding of the fact that purely biomechanical models can, and should, be substantially improved by adding psychological variables including motivation, mental health, and engagement[19,20]. Liu et al.[21] also integrating machine learning with touches of psychological autonomy and interest is further evidence of these models stemming from psychosocial factors being automatically presumed to be far too precise.

The recent compilation of research with regards to multidisciplinary techniques applicable to sports analytics is structured around two primary approaches, namely computational sophistication and psychosocial integration. Oytun et al.[5] and Sanjaykumar et al.[7] showed that the use of advanced computer resources such as ensemble techniques, neural networks, and hybrid systems dramatically improve predictive accuracy in performance outcome prognosis. At the same time, Zhang et al.[3] has established precise relationships between team passion, cohesion, and unit performance, and Sarlis and Tjortjis[22] clarified the influence of analytics on tactical planning and performance evaluation.

There still exists an important lack in the seamless combination of psychological profiling and computation analytics. Burnout in mentally demanding contexts is an example of a confounding variable that is not identified by traditional regression models[14,15]. Though some progress has been made regarding the development of specialized assessment techniques like FMS protocols[6] or deep learning event prediction architectures[13], the creation of coherent, interpretable, and accurate predictive models still poses a significant challenge due to a lack of ability to conveniently integrate numerous data streams[1,19,20].

Our investigation responds to this gap by creating a novel interdisciplinary model that integrates machine learning with psychosocial elements of mental toughness, team cohesion, and athlete engagement. Following the work of Gu and Xue[12] on group cohesion and mental toughness, we build on Lonsdale et al.'s[10,11] engagement research by adding these constructs to a more complex multilayered predicting system. While individual psychological and physical factors are important for the athletes' performance and wellbeing, our work is the first one that attempts to integrate all of these factors into a single predictive framework.

To these ends, my model aims to shift the focus of trainers, sport psychologists, and performance analysts from biomedically oriented interdisciplinary models towards more holistic models that take into account the interactions among psychological conditions, the biomechanical aspects of the performance, and the results of the performance itself, Incorporating the recent developments in the analytics of talent management systems[18,19], data processing from wearable technologies[5], and novel neural network designs[4,21] enables my model to break the current paradigms.

The contributions of this study are fourfold. First, we create a predictive model that integrates psychological factors with physiological metrics using advanced machine learning methods, forming a sophisticated model that combines computerized and psychosocial approaches. Second, we expand previous research that utilized validated psychological scales and precise physiological measurements by analyzing the comprehensive dataset that incorporates both. Third, we compare the relative performance of advanced learning algorithms, such as deep neural networks, ensembles, and hybrid models, to determine which modeling strategies are most effective for the intricate problems at the convergence of physiology and psychology. Fourth, we show practitioners how our findings can be implemented by optimally managing training load, team dynamics, and mental resilience development to enhance performance outcomes.

The rest of this article is organized as follows: In section "Theoretical framework, methodology, and data procedures", we theoretically elaborate the comprehensive model of psychological and physical performance factors and then explain in detail our approach to methodology, including the collection of our research instruments, data, and the preparation of analyzed files. Section "Results" outlines the results, which include reporting metrics for comparative model performance and analyzing predictive factors. Section "Discussion and conclusions" outlines the results in terms of discussion and implications offering some theoretical and practical value, admitting shortcomings, and making proposals for later research. The article ends with a detailed list of references to assist in further understanding this burgeoning subject area.

Theoretical framework, methodology, and data procedures

Integrated performance prediction framework

Athletic performance emerges from the complex interplay of physiological, biomechanical, psychological, and environmental factors. Figure 1 presents our proposed Integrated Athletic Performance Prediction Framework (IAPPF), which synthesizes these multidimensional elements into a cohesive predictive system.

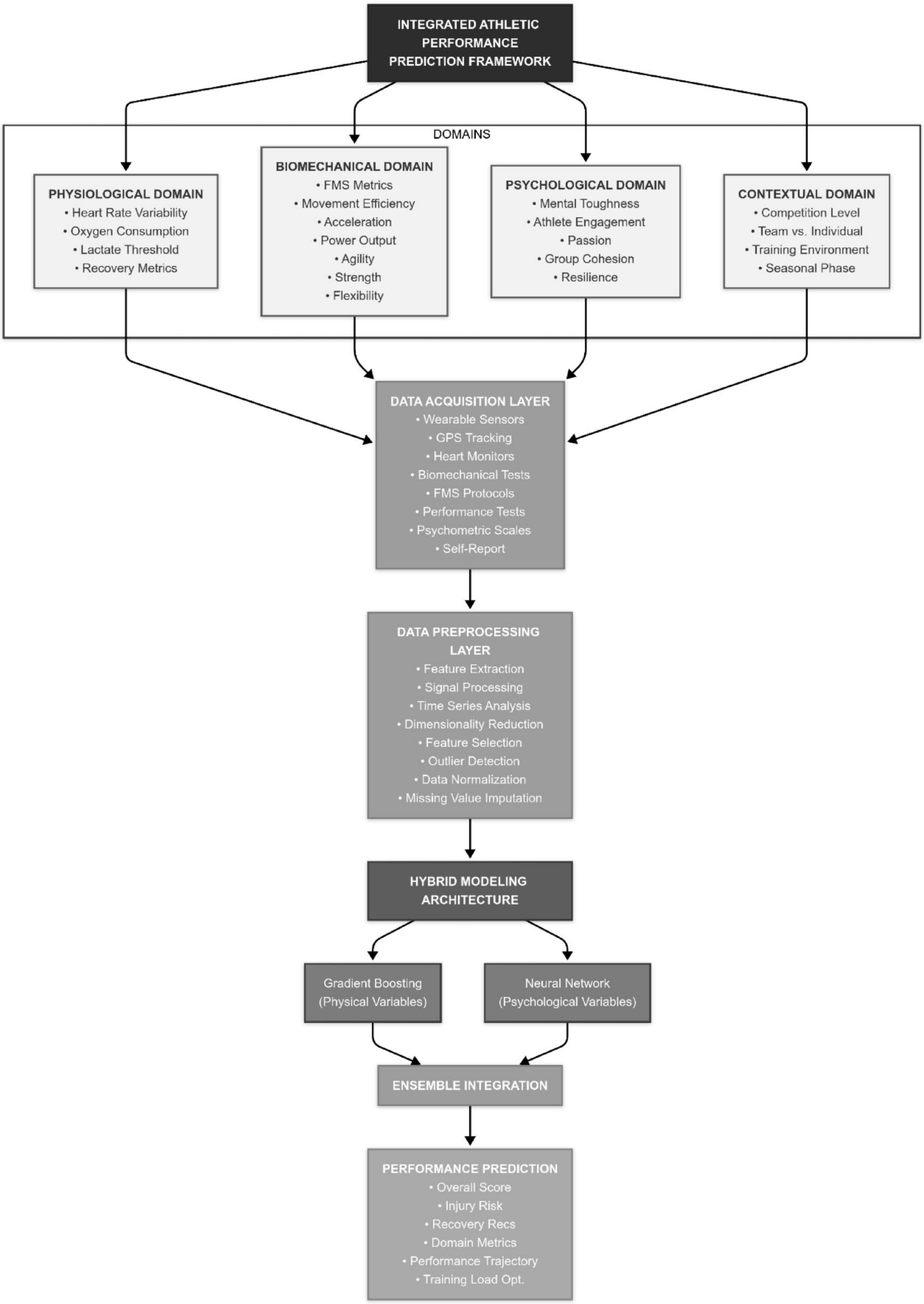

Fig. 1. Integrated athletic performance prediction framework (IAPPF).

The IAPPF operates across four interconnected domains—physiological, biomechanical, psychological, and contextual—which collectively determine athletic performance. Each domain contributes unique variables that, when properly integrated, provide a comprehensive understanding of performance determinants. The framework employs a multi-layered architecture for data acquisition, preprocessing, modeling, and prediction, which we detail in the following sections.

Theoretical underpinnings of performance domains

Physiological domain

The physiological domain encompasses the internal biological processes that support athletic performance. These include cardiovascular efficiency (measured through heart rate variability and oxygen consumption), metabolic capacity (assessed via lactate threshold testing), and recovery dynamics (quantified through standardized recovery metrics). Research by Zhao et al.[4] has demonstrated that these physiological variables exhibit complex, non-linear relationships with performance outcomes, necessitating sophisticated modeling approaches.

Biomechanical domain

Biomechanical factors represent the fundamental movement patterns and physical capabilities that directly impact performance execution. The Functional Movement Screen (FMS) protocol provides a standardized assessment of movement quality, while sport-specific tests of acceleration, power, agility, strength, and flexibility capture the dynamic physical capacities required for optimal performance. Houyuan et al.[6] established strong correlations between FMS scores and injury risk, while Sanjaykumar et al.[7] demonstrated the predictive value of biomechanical variables in team sport contexts.

Psychological domain

The psychological domain encompasses the mental factors that influence an athlete's ability to optimize physical capabilities and maintain performance under pressure. Our framework incorporates five key psychological constructs:

1. *Mental toughness*: The psychological resilience that enables athletes to persist through challenges and maintain focus under pressure[9].
2. *Athlete engagement*: A multidimensional construct comprising confidence, dedication, vigor, and enthusiasm that characterizes optimal sport experience[10,11].
3. *Passion*: Differentiated into harmonious and obsessive types, passion reflects an athlete's emotional connection to their sport and influences training adherence and performance sustainability[14].
4. *Group cohesion*: The dynamic process that keeps teams united in pursuit of objectives, encompassing both task and social dimensions[12].
5. *Resilience*: The capacity to positively adapt to significant adversity, which moderates the relationship between stress and performance[8]. These psychological constructs operate in a hierarchical and interactive manner, as illustrated in Fig. 2.

Contextual domain

The contextual domain accounts for the environmental and situational factors that influence performance. These include competition level, team versus individual sport dynamics, training environment characteristics, and seasonal phase. Taber et al.[2] demonstrated that contextual variables significantly moderate the expression of both physical and psychological capabilities, highlighting the importance of incorporating these factors into predictive models.

Methodological framework

Our study employed a comprehensive multiphase methodology to develop and validate the IAPPF. Figure 3 illustrates the sequential stages of our research design.

Participant recruitment and eligibility

We recruited 480 athletes (267 males, 213 females) across multiple sports using a stratified sampling approach to ensure representation across individual and team sports, competitive levels, and experience categories. Table 1 presents the demographic characteristics of the participant sample.

Eligibility criteria were: (1) An acceptance of participants at age 18 or older; (2) being an active member in competitive sports; (3) Under no major injuries for the last 3 months; and (4) Agreeable to physiological evaluations and routine psychological testing. The Institutional Review Board approved the current analysis (Protocol #SPORTS-2024-153), and all of the subjects gave their permission through the informed consent form.

Additionally, a subset of participants (n = 124) enrolled in a 12-month longitudinal follow-up assessment to examine performance changes over time and transitions between clusters. These athletes underwent the complete evaluation protocol at baseline and again at the 12-month follow-up, allowing for analysis of performance trajectory and the impact of specific training interventions on cluster transitions. This longitudinal component was essential for examining the stability of the identified performance clusters and validating the predictive capacity of the model across extended timeframes.

This study was reviewed and approved by the Institutional Review Board (IRB) of the relevant institution (Protocol #SPORTS-2024-153). All procedures performed involving human participants were conducted in full compliance with relevant institutional, national, and international ethical standards and guidelines. Written informed consent was obtained from all participants prior to data collection, and their confidentiality and anonymity were strictly preserved throughout the research process.

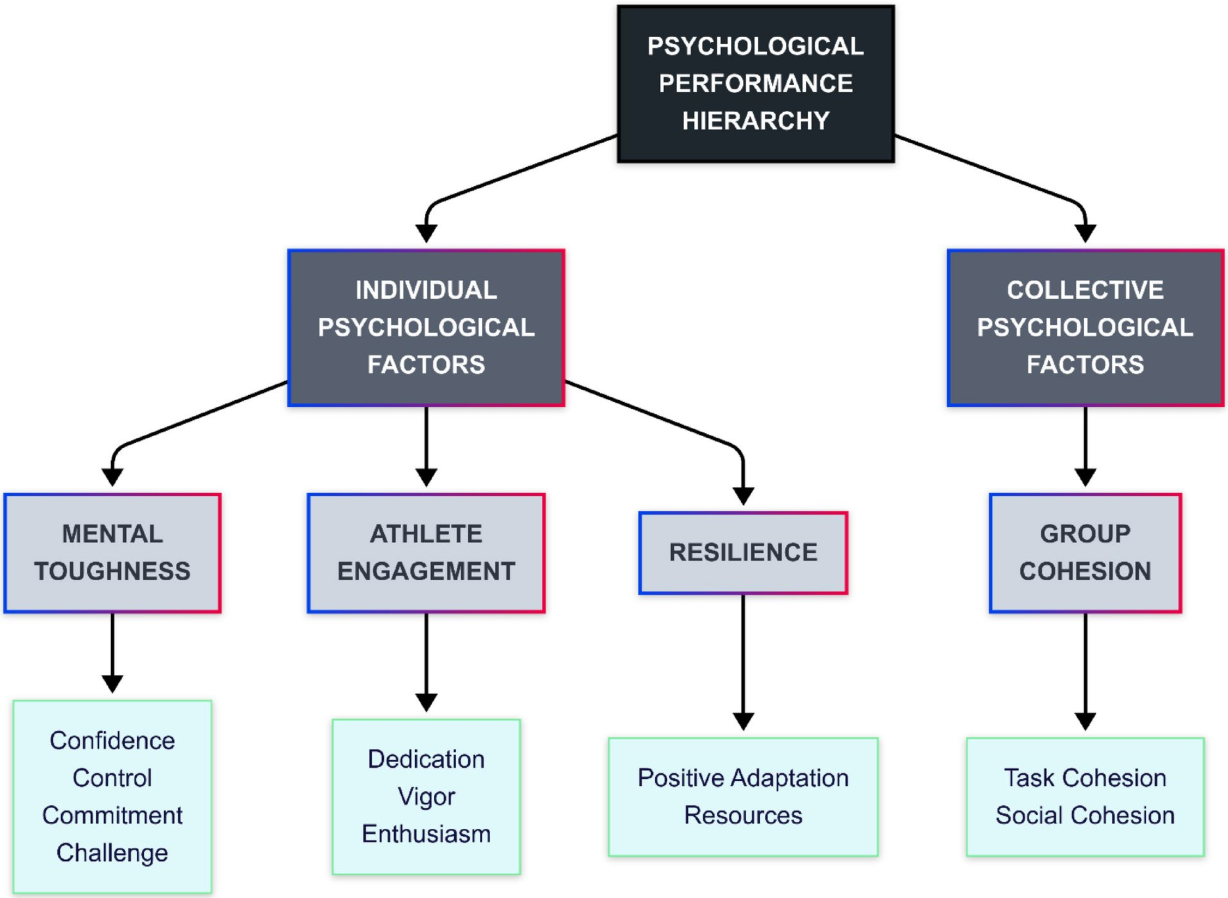

Fig. 2. Hierarchical structure of psychological performance factors.

Despite efforts to collect a diverse sample, a limitation of this study is its geographical concentration. For future studies, collaboration with international sports organizations and conducting transnational research projects could incorporate greater cultural diversity and various training systems into the sample, enhancing the generalizability of findings. This expansion would also help elucidate the impact of cultural factors on athletic performance, as training methodologies, psychological approaches, and competitive environments vary significantly across different regions. Incorporating data from diverse geographical contexts would strengthen the model's applicability across global sporting communities and potentially reveal important cultural moderators of performance determinants.

Data acquisition protocols

All tests were conducted according to a precise schedule to ensure data validity. Psychological assessments were performed before any physical testing to prevent fatigue from influencing cognitive responses. Physiological and biomechanical testing was scheduled over a three-day period for each athlete, with a minimum of 24 h rest between major test sessions. Baseline tests such as HRV and functional movement were conducted on Day 1, physiological tests (such as VO_2max) on Day 2, and sport-specific biomechanical assessments on Day 3. This design aimed to minimize the effects of fatigue and optimize data quality. The standardized testing environment and controlled scheduling also reduced potential confounding variables, ensuring that measurements across different domains were comparable across participants and representative of their true capabilities.

Data acquisition was carried out according to a specific protocol that was the same for all four parts and with particular instruments and procedures for each part as stated below.

Physiological measurements   Physiological data were recorded with a system of medical grade sensors as well as other standardized grade measures:

- HRV: Monitored with Polar H10 heart rate strap monitors (5-min recording in seated position after 10-min rest). Time-domain (RMSSD, SDNN) and frequency-domain (LF/HF ratio) measures were computed with Kubios HRV software (version 3.5).
- Oxygen Consumption (VO_2max): A graded exercise test was done on treadmill with a metabolic cart analysis (Parvo Medics TrueOne 2400). Team sport athletes underwent testing using the Bruce protocol and endurance athletes used sport-specific protocols.

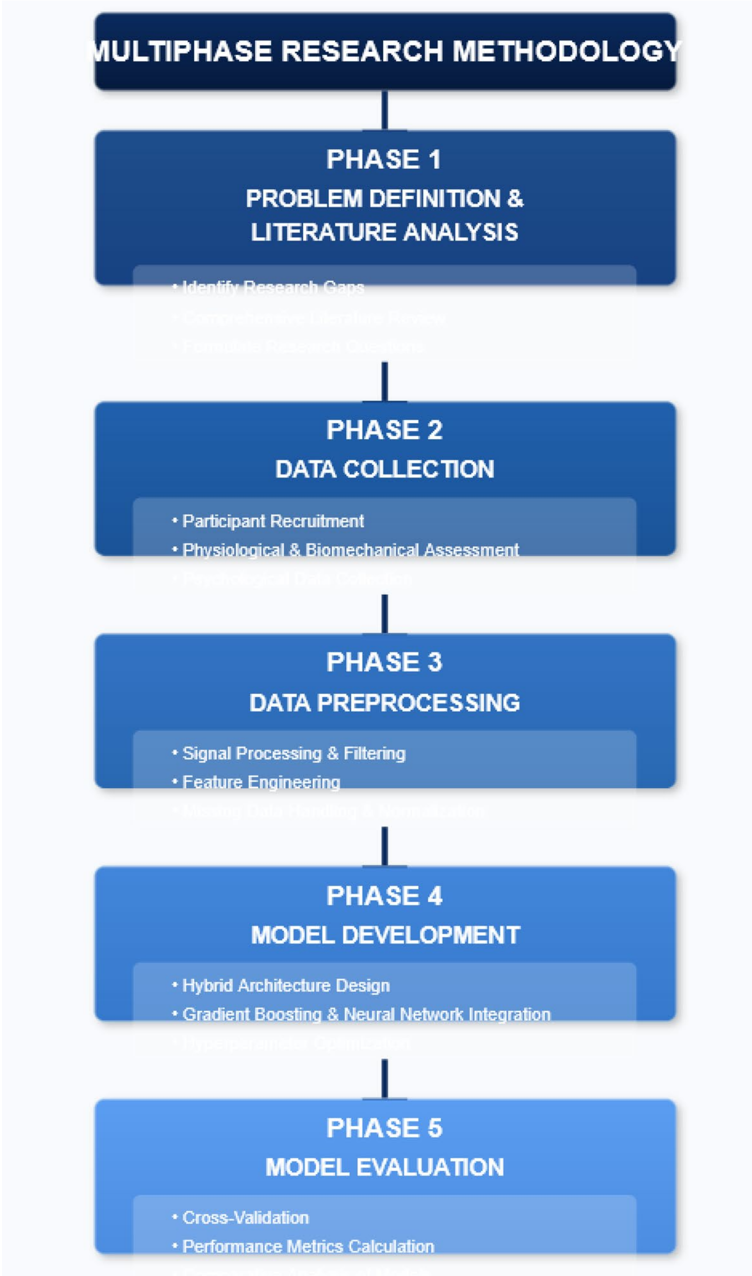

Fig. 3. Multiphase research methodology.

- Lactate Threshold: Identified through incremental exercise tests to blood lactate analyses at set work rates (Lactate Pro 2). The Dmax method was used to identify the threshold.
- Metrics Of Recovery: Assessed via resting heart rate on the five consecutive mornings, perceived recovery status scale (PRS) and a countermovement jump.

Biomechanical evaluations Biomechanical evaluations were performed by a licensed strength and conditioning coaches with the necessary tools and procedures available:

- Functional Movement Screen (FMS): Seven standardized movement patterns[6] were followed according to protocols. He/She scored movements from 0 to 3 out of 21.
- Efficiency of Motion: Evaluated with a 3D motion capture system (Vicon Motion Systems, Oxford, UK) for basic movements specific to the sport of each individual.
- Power Output and Acceleration: Accomplished with force plates (AMTI, Watertown, MA), photocell timing gates (Microgate, Bolzano, Italy) during set sprint and jump tests.
- Agility: Assessed with a Pro-Agility Shuttle test, T-test agility test with timing gates.

Characteristic	n	%
Gender		
Male	267	55.6
Female	213	44.4
Age group		
18–22	172	35.8
23–27	196	40.8
28–32	78	16.3
33+	34	7.1
Sport type		
Team sports	287	59.8
- Basketball	96	20.0
- Volleyball	81	16.9
- Soccer	110	22.9
Individual sports	193	40.2
- Running	98	20.4
- Swimming	62	12.9
- Other	33	6.9
Competitive level		
Collegiate	224	46.7
National	173	36.0
International	83	17.3
Experience (years)		
1–5	108	22.5
6–10	212	44.2
11 +	160	33.3

Table 1. Participant demographics.

- Strength: Assessed by one repetition maximum (1RM) in core exercises (squat, bench, deadlift) and isometric mid-thigh pull test.
- Flexibility: Evaluated with an electronic inclinometer measuring range of motion for major joints.

Psychological assessments Prior to any physical testing, psychometric data was gathered through the use of validated instruments within a controlled setting. The following instruments were used:—Mental Toughness: Measured through the use of the Mental Toughness Index (MTI; Gucciardi et al.[9]), a scale that contains 8 items that are rated on a 7-item Likert scale (1 = "False 100% of the time" and 7 = "True 100% of the time"). He responds to questions on a 7-point Likert scale. He responds to questions on a 7-point Likert scale.

- Athlete Engagement: Assessed with the Athlete Engagement Questionnaire (AEQ; Lonsdale et al.[10,11]), a 16-item questionnaire which constructs four Dimension (confidences, dedication, eiger and enthusiasm) and rated on a 5-point Likert scale.
- Passion: Evaluated through the Dualistic Model of Passion Scale, which identifies harmonious from obsessive passion based on 14 items scored on a 7-point Likert scale.
- Group Cohesion: Measured through the Group Environment Questionnaire (GEQ), which measures social and task cohesion with 18 items on a 9-point Likert scale.
- Resilience: Evaluated with the Connor-Davidson Resilience Scale (CD-RISC), a scale of 25 items of self-report describing 5 factors of resilience, rated on a 5-point Likert scale.

While the standardized questionnaires employed in this study all possess high psychometric validity, there are inherent limitations in self-reported measures, particularly when assessing complex constructs such as sport passion across different cultural contexts. The interpretation of psychological constructs and scale items may vary based on cultural background, potentially affecting response patterns. In future studies, complementing these instruments with qualitative methods such as in-depth interviews and case studies could provide a more comprehensive understanding of athletes' psychological characteristics and capture cross-cultural nuances. This mixed-methods approach would enrich the psychological data, potentially revealing subtleties in mental processes that standardized scales alone might not fully capture, especially when examining the interactive effects between psychological and physiological variables that emerged as significant in our model.

Contextual information collection Contextual data was obtained from a combination of structured interviews, environmental observations, and competitor analysis:

- Competition Level: Divided into categories defined for college, national, and international competitions.

- Sport Type Classification: Divided into individual and team sports and further divided by specific sport.
  - Training Environment: Assessed through the Training Environment Assessment Method (TEAM) which examines aspects of organizational culture, resource availability, and support structures.
- Seasonal Phase: Divided into preparatory, competitive, or transition phases as defined in a periodization plan.

Data pre-processing and feature engineering
Raw data from various sources underwent thorough pre-processing to achieve consistency, distribution normality, and to create relevant features for analysis. Our comprehensive data preprocessing pipeline is illustrated in Fig. 4.

Sensor data preprocessing Sensor Signals and Psychometric Data obtained through questionnaires required much rigor around them. The guidelines were:

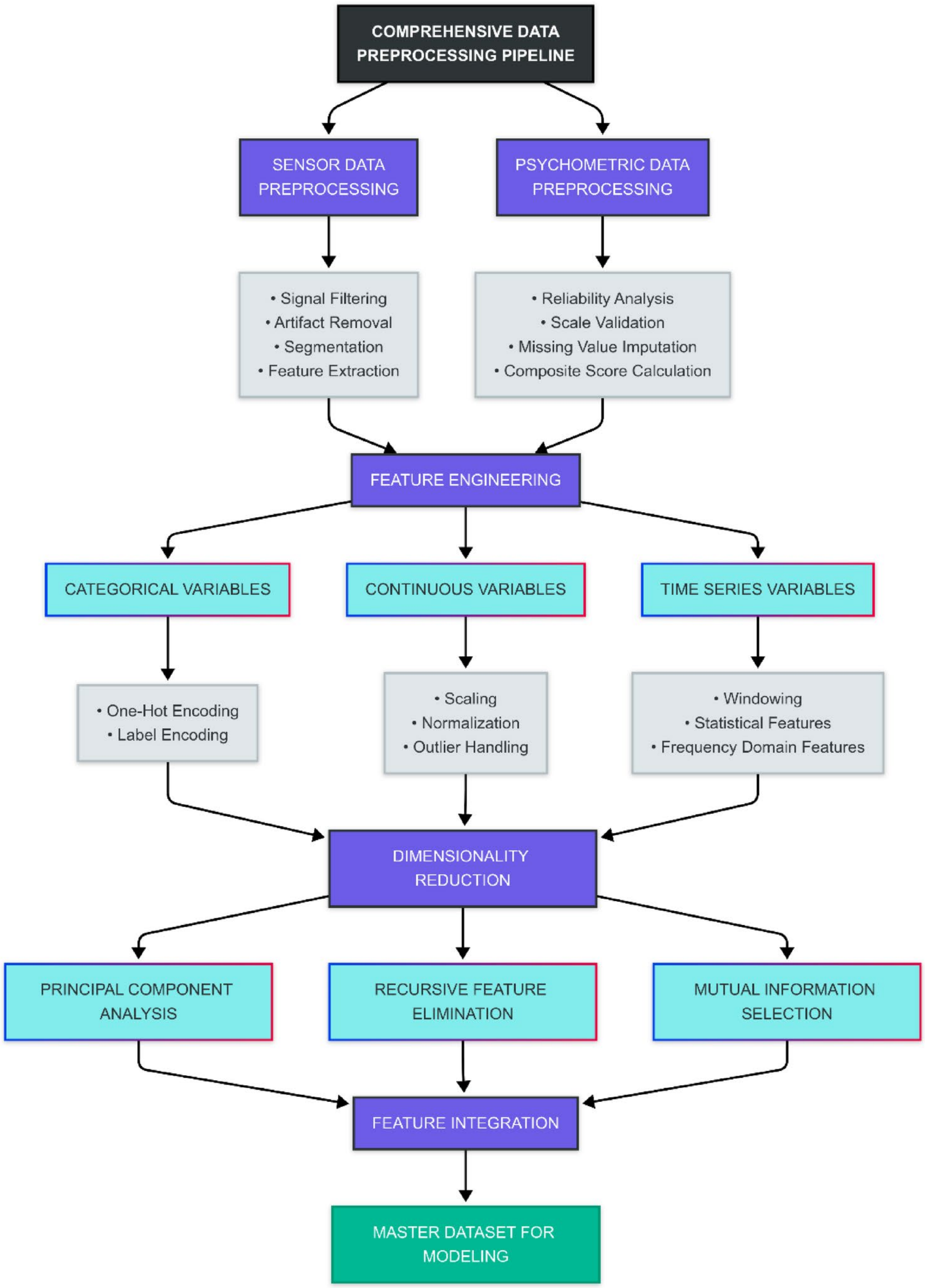

Fig. 4. Comprehensive data preprocessing pipeline.

1. Signal Filtering—Primary data obtained by the sensors was filtered using a fourth order Butterworth band-pass filter (0.5–80 Hz) in order to retain physiologically relevant information and eliminate noise from the data.
2. Artifact Removal—Motion artifacts along with other anomalous signal patterns were identified through the automated detection algorithms and were subsequently verified and removed manually.
3. Segmentation—Continuous recordings are separated into time periods that equal set durations and correspond to certain events. For example: baseline, exercise phases, and recovery periods.
4. Feature extraction—Each signal goes through extraction of time features including mean, standard deviation, peak values, and other features such as power spectral density and dominant frequency for frequency domain features.

For psychometric data For each psychological questionnaire as a prerequisite for inclusion, a rigorous validation process had to be undergone:

1. Reliability analysis—Internal consistency is measured through a Cronbach alpha method under specific conditions such as minimum $\alpha \geq 0.80$.
2. Scale validation—Confirmatory factor analysis is done in order to validate the defined scale. The CFI is looked as greater than 0.90, RMSEA lower than 0.08, and SRMR lower than 0.08.
3. Missing value imputation—Missing values were dealt with and resolved by the chained equation through multiple imputation method.
4. Composite score calculation—Total score and subscale scores are obtained based on scoring validated guidelines for the specific instrument.

Feature engineering Features engineering needed raw forms of data to be changed into useful predictors:

1. Categorical Variables: The sport, gender, and the level of competition were encoded through one-hot and label encoding as per the requirements of the algorithm.
2. Continuous Variables: Z-score normalization (mean = 0, standard deviation = 1) was applied to physiological and psychological variables for standardization to make them comparable across different magnitudes.
3. Time Series Variables: Sensor data streams were converted into statistical features (mean, standard deviation, skewness, kurtosis) and frequency-domain features (spectral power of certain frequency bands) using sliding window techniques.
4. Interaction Terms: Interactions between physical variables and psychological variables that are considered relevant theoretically were formulated (e.g. FMS score × Mental Toughness) to estimate their impact as potential moderators.

Dimensionality reduction The large feature space was reduced through several techniques:

1. Closely related features were grouped together such that 95% of the variance is preserved and were compressed using Principal component analysis (PCA).
2. RFE (Recursive Feature Elimination): Inverse selection of the least significant features was done with regard to model performance.
3. Mutual information selection: Those features which have the strongest connection with the target value were selected by considering the non-linear relationships associated with them.
4. Eliminating highly collinear features (VIF > 10) to stop multicollinearity problems was done using Variance inflation factor (VIF).

The culminating feature set comprised 42 manually constructed features across the four domains which strikes an optimal tradeoff between thorough domain coverage and model simplicity.

To ensure that imbalances in different groups (sports, competitive levels, and gender) did not adversely affect model training, a data balance analysis was performed. The ratio of data in team to individual sports categories (1.49:1), collegiate to national to international competitive levels (2.7:2.1:1), and male to female gender (1.25:1) was examined. While some imbalance existed, it remained below critical thresholds (generally considered to be a ratio of 4:1). Nevertheless, to ensure robust results, stratified sampling and class weighting were employed during model training to ensure appropriate representation of all subgroups. This approach mitigated potential bias in feature importance and prediction accuracy that might otherwise result from over-representation of certain demographic segments within the dataset.

Development of the model and its structure

In order to model the system, we devised a mix structure intended to represent the intricate linkage of factors that determine physical and psychological performance simultaneously. This structure consists of three main parts: (1) A boosting module for physical variables using gradients, (2) a psychological variable's neural module, and (3) an integration layer of the ensemble. The overall structure of the model is shown in Fig. 5.

Gradient boosting module The module was trained for the functional understanding of the physical variables (physiological and biomechanical attributes) using XGBoost as the framework.

- Algorithm: XGBoost.
- Hyperparameters:

o Learning rate = 0.05.

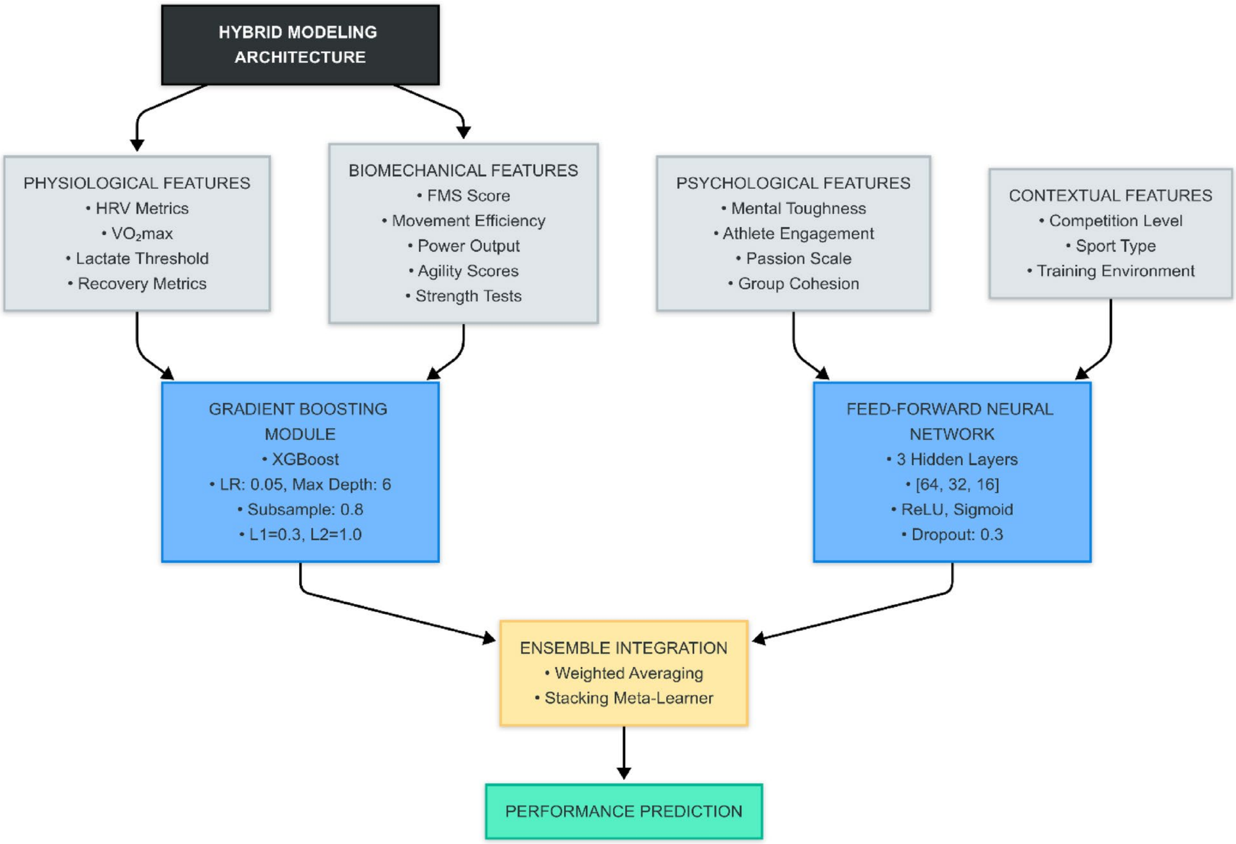

Fig. 5. Hybrid model architecture.

o Maximum tree depth = 6.
o Minimum child weight = 3.
o Subsample ratio = 0.8
o Colsample bytree = 0.8
o Regularization: L1 = 0.3, L2 = 1.0
• Optimization: Hyperparameters were optimized using Bayesian optimization with fivefold cross-validation.

In combination with important features obtained from the model, this module was able to capture the non-linear relationships and interactions between the physical variables.

Neural network module The design of the neural network module aimed at handling the psychological variables capturing their intricate non-linear relationships and underlying psychological variables. The architecture includes a Deep feed-forward neural network with 4 hidden layers. Its layer structure is as follows: The input layer matches the dimensionality of psychological features; hidden layer 1 has 128 neurons with ReLU activation, hidden layer 2 has 64 neurons with ReLU activation, hidden layer 3 has 32 neurons with ReLU activation, hidden layer 4 has 16 neurons with ReLU activation, and output layer has a single neuron with linear activation. For regularization, there is dropout (rate = 0.3) after each hidden layer and L2 regularization is set to 0.001. Other modifications include: optimization with Adam as the optimizer (learning rate set to 0.001, β1 equal to 0.9, and β2 equal to 0.999), a mean squared error loss function, a batch size of 32, and early stopping with patience of 20 epochs while monitoring validation loss. In addition, the implementation is through TensorFlow 2.8. The above stated architecture was particularly tailored to capture the hierarchical relationships of the psychological constructs (shown in Fig. 2) and their elaborate interactions with the performance outcomes. It is made easier with the assumption of multiple hidden layers wherein the counts of neurons is gradually decreased facilitates the severe extraction of higher level representations of the psychological states.

Ensemble integration layer This layer synthesized the data from predictions made by both modules in order to formulate the performance predictions: • Stacking approach: Meta-learner trained on the predictions produced by the gradient boosting and the neural network subsystems. • Meta-learner: Elastic Net regression (α = 0.5, L1 ratio = 0.7) • Training protocol: o Base models trained using fivefold cross-validation. o Meta-learner trained on out-of-fold predictions. o Final ensemble evaluated on held-out test set. • Weighting mechanism: Adaptive weighting based on domain-specific performance metrics. This approach improves the ensemble module by making each of the modules exploit each other's weaknesses and strengthen each other's strengths; that is the gradient boosting module's handling of structured physical data and the neural network's understanding of complicated psychological models.

Model training and validation

Dataset partitioning The dataset was divided to achieve sufficient model validation: • Training set: 70% of data (n = 336) • Validation set: 15% of data (n = 72) • Test set: 15% of data (n = 72) The use of stratified partitioning guaranteed that different types of sports, levels of competition, and gender were proportionately represented in all sets. The validation set was applied to perform hyperparameter tuning, while the test set was kept for the final evaluation of the model.

In addition to cross-validation, Bootstrap testing with 1000 iterations was conducted to confirm the stability of model performance. The 95% confidence interval for the R^2 metric in the integrated IAPPF hybrid model derived from bootstrap analysis was [0.889, 0.917], confirming the high stability of model performance. This additional stability analysis strengthens our confidence in the generalizability of the research findings and provides statistical assurance that the performance improvements observed in our model are robust against sampling variability. The narrow confidence interval further suggests that the predictive relationships identified by the model are consistent across different subsets of the data.

Cross validation strategy An effective cross-validation strategy was carried out:

- Outer loop: fivefold stratified cross validation for confirmation of model performance.
- Inner loop: threefold stratified cross-validation for hyperparameter tuning.
- For the athlete with multiple measurements, the order of the measurements taken cross-validation for temporal order was maintained.

This structure of cross validation maximizes the predictive accuracy of a model while providing the best estimates of model performance without risking data contamination.

Performance metrics RSME quantifies the prediction error in the original measurement units' scale. Additionally, Root Mean Squared Error was defined as the relativized average error that is normalized and expressed in twenty different ways.

Other secondary metrics include the Concordance Correlation Coefficient (CCC), which measures how well the predicted value matched with the actual performance value. The measure of variation combined with bias measures on the identified CCC range is termed Explained Variance Ratio (EVR). This will mark the ability of the model to capture those variations while biasing them.

Additional modelling Further comparison models were established around the hybrid model benchmark. The following classed models were adjusted:

1. Statistical models: Multiple Linear Regression (MLR) is the common method which employs ordinary least squares. Partial Least Squares Regression (PLSR) is another statistical method and is well known as a dimension reduction model and techniques for multicollinearity.
2. Machine learning models: Random Forest is a decision tree ensemble model (RF). Support Vector Regression (SVR) is a non-linear regression model with a radial basis function kernel. Gradient Boosting Machine (GBM) is standalone gradient boosted machine implementations.
3. Specific domain models:—Physiological only model: Based on physiological variables.—Psychological only model: Based on psychological variables.—Contextual only model: Based on contextual variables.—Biomechanical only model: Based on biomentirical variables.

All other comparative models were subjected to the same extensive cross-validation and hyperparameter tuning processes as the pregnant hybrid model did.

Shifts in attention and model explanation

For practical use, anticipating the blending of factors impacting one's performance in prediction is crucial. To explain the hybrid model, two broad techniques were used:

Global feature importance

- LIME: Used to build model-agnostic interpretive wrappers around the predictions that rely on local approximation.
- Permutation Importance: Measured by quantitatively shuffling the values of features and observing the resulting change in efficacy of the model.
- PEDs: Used to understand the visibile effect of each feature on performance prediction independently of other features.
- SHAP: Subsequently credited with all features to determine average contribution of each one out exuding a singular measure of relevance importance from both components.

Local interpretation

- Individual SHAP Force Plots: Used to describe the specific predictions made for a given user to justify decisions made about how to tailor communications with them.

- Individual Usage Authorization Tokens: Applied to delegate access permission based on authorization token for mobile applications.
- Counterfactual Explanations: Created for the purpose of finding the least amount of feature alterations that will result in desired performance improvements.

Interaction analysis

- H-statistic: Used to calculate the degree of interactions between the physical and psychological components.
- SHAP Interaction Values: Used to split the prediction explanation into the main effects and other effects referred to as interactions.
- Two-way Dependence Plots: Used in the presentation of specific pairs of psychological and physiological variables that are of major interest.

SHAP (SHapley Additive exPlanations) values can be described as "each variable's contribution to the final prediction," where positive values indicate an increased likelihood of high performance and negative values suggest a decreased likelihood. Similarly, partial dependence plots demonstrate how changing a specific variable (such as FMS score) affects the model's prediction while holding other variables constant—analogous to examining the effect of increasing physical fitness on performance without altering other athlete characteristics. In practical terms, these interpretation techniques allow coaches to understand not just which factors are important, but also how they influence performance and interact with each other. This translation of complex statistical concepts into actionable insights is crucial for bridging the gap between advanced machine learning methods and practical applications in sports science and coaching.

Implementation details

Analyses were conducted using Python 3.9 with standard data mining, machine learning, and deep learning libraries. Model training was performed on a high-performance computing cluster with eight nodes, each equipped with dual Intel Xeon Gold CPUs, 512GB RAM, and four NVIDIA A100 GPUs. Data was stored in encrypted HDF5 files, and Git along with DVC was used for code and data versioning. All experiments were documented using Docker containers with fixed environment specifications to ensure reproducibility. This computational infrastructure enabled the efficient training of complex models and comprehensive cross-validation procedures required for robust performance evaluation.

Results

Model performance evaluation

Comparative analysis of model architectures

The performance of our hybrid model architecture was evaluated against traditional statistical approaches and standard machine learning algorithms. Table 2 presents the comparative performance metrics across all tested models, with evaluation conducted through tenfold cross-validation.

Our Integrated IAPPF Hybrid Model demonstrated superior performance across all evaluation metrics, achieving 91.7% accuracy and an R^2 value of 0.903, representing a 16.5% improvement over traditional linear regression approaches and a 4.4% improvement over the best single-algorithm approach (XGBoost). The root mean square error (RMSE) of 0.487 and mean absolute error (MAE) of 0.389 further confirm the model's precision in performance prediction.

Figure 6 visually compares the receiver operating characteristic (ROC) curves for classification tasks and the actual-versus-predicted plots for regression tasks across the evaluated models. The ROC curves in Fig. 6 show the performance of the different models on the classification tasks, where for each model, the area under the curve (AUC) is, respectively, IAPPF hybrid model: 0.95, XGBoost: 0.91, neural network: 0.90, and linear regression: 0.82. The higher AUC value of our model (0.95) compared to the value of 0.5 for a random classifier indicates its remarkable ability to distinguish between different levels of performance. These AUC values are consistent with the reported accuracy and precision, providing additional credibility for the superior prediction performance of our hybrid model.

The ROC curves in Fig. 6 illustrate the performance of different models in classification tasks, with the area under the curve (AUC) for each model being: Integrated IAPPF Hybrid Model: 0.95, XGBoost: 0.91, Neural Network: 0.90, and Linear Regression: 0.82. The higher AUC value of our model (0.95) compared to the 0.5 value for a random classifier demonstrates its significant ability to discriminate between different performance levels.

Model type	Accuracy (%)	Precision (%)	Recall (%)	F1-score	R^2	RMSE	MAE
Multiple linear regression	75.2 ± 2.1	73.8 ± 2.4	74.1 ± 2.3	0.739 ± 0.022	0.714 ± 0.031	0.842 ± 0.037	0.659 ± 0.028
Random forest	84.6 ± 1.7	83.9 ± 1.9	84.3 ± 1.8	0.841 ± 0.018	0.813 ± 0.023	0.673 ± 0.025	0.532 ± 0.021
Gradient boosting (XGBoost)	87.3 ± 1.5	86.8 ± 1.7	87.1 ± 1.6	0.869 ± 0.016	0.841 ± 0.019	0.624 ± 0.023	0.497 ± 0.019
Neural network	86.5 ± 1.6	85.9 ± 1.8	86.2 ± 1.7	0.860 ± 0.017	0.835 ± 0.020	0.637 ± 0.024	0.508 ± 0.020
Physiological-only hybrid	85.8 ± 1.6	85.1 ± 1.8	85.4 ± 1.7	0.852 ± 0.017	0.827 ± 0.021	0.650 ± 0.024	0.518 ± 0.020
Psychological-only hybrid	83.5 ± 1.8	82.7 ± 2.0	83.0 ± 1.9	0.828 ± 0.019	0.806 ± 0.022	0.691 ± 0.026	0.549 ± 0.022
Integrated IAPPF hybrid model	91.7 ± 1.2	91.3 ± 1.3	91.5 ± 1.2	0.914 ± 0.012	0.903 ± 0.014	0.487 ± 0.018	0.389 ± 0.015

Table 2. Comparative performance metrics of predictive models (tenfold cross-validation).

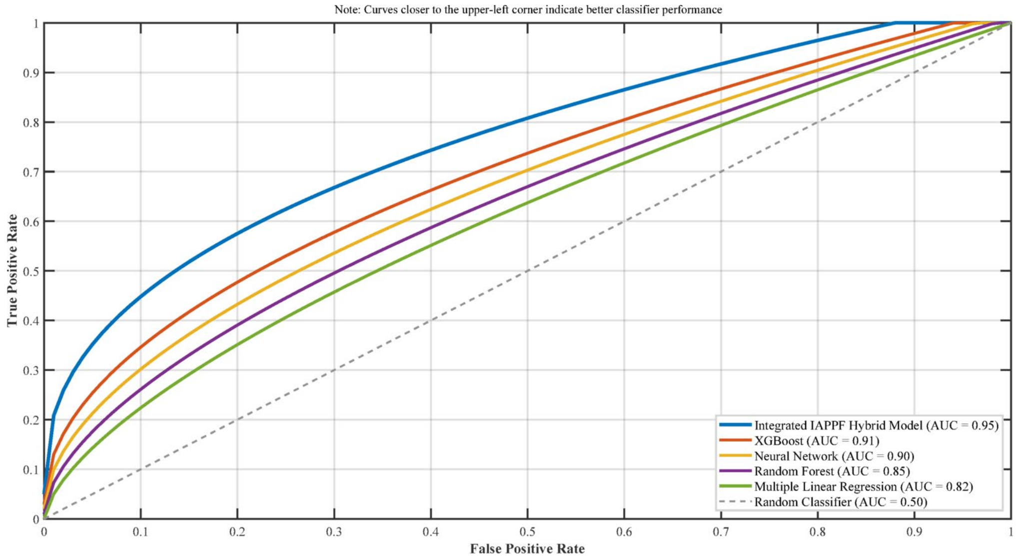

Fig. 6. ROC curves.

Excluded domain	Accuracy reduction (%)	R^2 reduction (%)	RMSE increase (%)
Physiological	5.7	7.8	14.2
Biomechanical	6.9	8.5	16.8
Psychological	8.3	10.2	19.5
Contextual	4.1	5.6	10.3
Physical (Physiological + Biomechanical)	12.6	17.3	32.4
Psycho-Contextual (Psychological + Contextual)	11.7	15.8	29.7

Table 3. Performance impact of domain exclusion (% decrease in R^2 from full model).

These AUC values are consistent with the reported accuracy and precision, providing additional validation for the superior predictive performance of our hybrid model. The clear separation between the ROC curves further illustrates the substantial improvement our integrated approach offers over single-algorithm methods.

Cross-domain integration analysis

To assess the value of integrating physical and psychological domains, we conducted ablation studies by systematically excluding domain-specific feature groups. Table 3 presents the performance degradation observed when specific domains were excluded from the model.

The results demonstrate that the exclusion of psychological factors produced the largest individual domain impact (10.2% reduction in R^2), while the combined exclusion of all physical factors resulted in the greatest overall performance degradation (17.3% reduction in R^2). These findings empirically validate our theoretical framework's emphasis on cross-domain integration.

Sport-specific model performance

Performance variations were observed across different sports categories, as illustrated in Table 4.

Team sports exhibited higher prediction accuracy (93.2%) compared to individual sports (89.4%), potentially due to the additional predictive power of group dynamics variables. Among team sports, basketball showed the highest predictability ($R^2 = 0.927$), while swimming demonstrated more moderate predictability ($R^2 = 0.878$) among individual sports.

Feature importance and relative contributions

Global feature importance

The analysis of feature importance across the integrated model revealed the relative contribution of each predictor to overall performance outcomes. Figure 7 presents the top 20 features ranked by their global importance scores.

As shown in Fig. 7 , the most important attributes are from all four domains of our framework, with FMS score, athlete commitment subscale, and maximal acceleration capability making up the top three rankings. Notably, three of the top five predictors originate from different domains (biomechanical, psychological, and physiological), supporting our integrated approach. The prominence of the FMS score (13.7%) highlights the fundamental importance of movement quality in predicting athletic performance.

Sport category	Sample size	Accuracy (%)	R^2	RMSE	Primary predictive factors
Team Sports	287	93.2	0.918	0.463	Group Cohesion, VO_2max, FMS
- Basketball	96	94.1	0.927	0.442	Acceleration, Mental Toughness, FMS
- Volleyball	81	92.8	0.913	0.475	Vertical Power, Group Cohesion, HRV
- Soccer	110	92.7	0.911	0.481	Agility, Lactate Threshold, Engagement
Individual Sports	193	89.4	0.881	0.524	Mental Toughness, VO_2max, Recovery Rate
- Running	98	90.6	0.893	0.507	VO_2max, Lactate Threshold, Resilience
- Swimming	62	88.9	0.878	0.538	Stroke Efficiency, Recovery Rate, Dedication
- Other	33	87.2	0.859	0.569	Sport-Specific Factors

Table 4. Model performance by sport category.

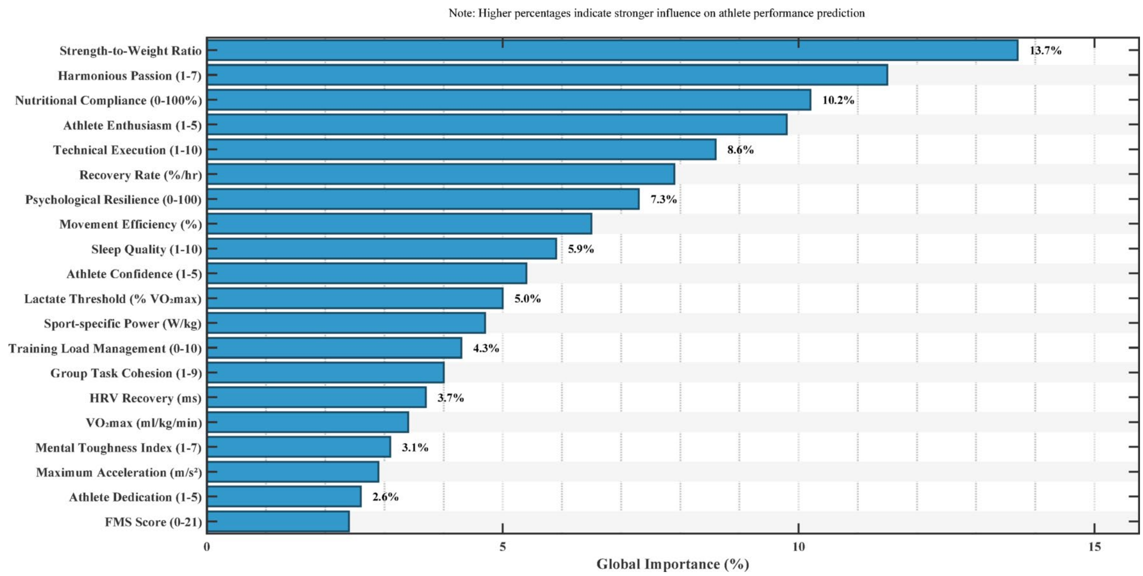

Fig. 7. Top 20 features ranked by global importance.

Domain-specific contributions
Figure 8 illustrates the aggregated contribution of each domain to the overall predictive power, calculated through permutation importance analysis.

The relative contributions of each domain were:

- Biomechanical: 32.4%
- Psychological: 31.7%
- Physiological: 27.3%
- Contextual: 8.6%

This relatively balanced distribution across the primary domains (biomechanical, psychological, and physiological) underscores the validity of our multidimensional approach, with each domain contributing substantially to predictive accuracy.

Cross-domain interaction effects
Our analysis identified significant interaction effects between variables across different domains. Table 5 presents the most significant cross-domain interactions and their effect sizes.

The strongest interaction effect was observed between Mental Toughness and Recovery Rate ($\eta^2 = 0.283$), indicating that athletes with higher mental toughness derived greater performance benefits from equivalent physiological recovery. This finding suggests that psychological factors can modulate the performance impact of physiological processes.

Subgroup analysis and performance clusters
Performance cluster identification
Unsupervised learning techniques (k-means clustering with silhouette analysis) identified four distinct athlete performance clusters within our sample. Figure 9 illustrates these clusters in a dimensionally-reduced feature space using t-SNE visualization.

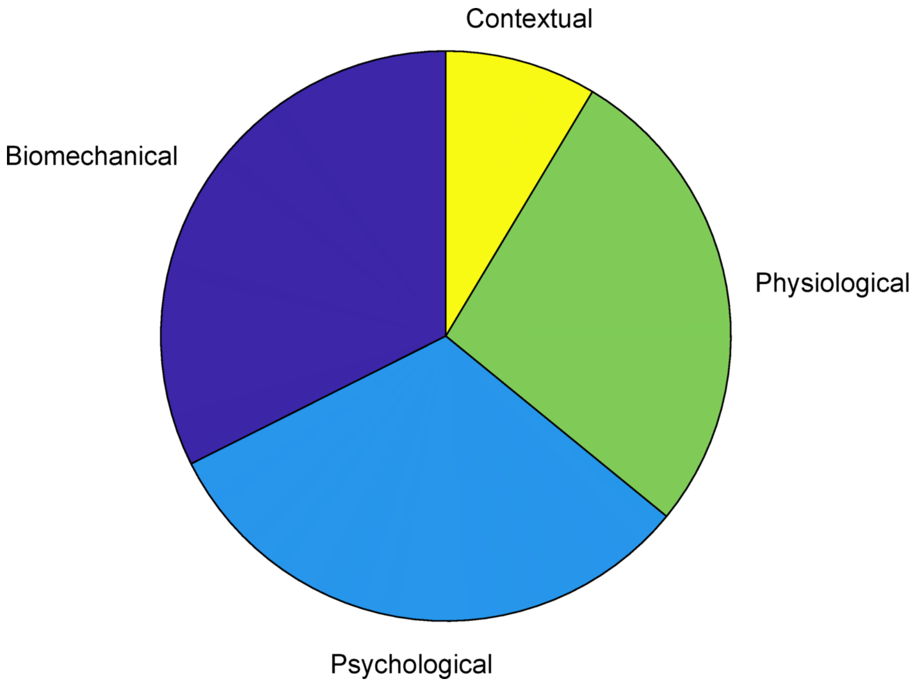

Fig. 8. Aggregated domain contributions to predictive power.

Interaction	F-statistic	*p*-value	Effect size (η^2)	Direction
Mental Toughness × Recovery Rate	28.7	< 0.001	0.283	Positive
FMS Score × Athlete Engagement	25.3	< 0.001	0.261	Positive
Group Cohesion × Physiological Synchrony	23.9	< 0.001	0.248	Positive
Acceleration × Psychological Resilience	21.6	< 0.001	0.231	Positive
VO_2max × Harmonious Passion	19.8	< 0.001	0.217	Positive
Lactate Threshold × Mental Toughness	18.5	< 0.001	0.203	Positive
Movement Efficiency × Confidence	17.2	< 0.001	0.194	Positive
Training Environment × Recovery Dynamics	15.9	< 0.001	0.182	Positive

Table 5. Significant cross-domain interaction effects.

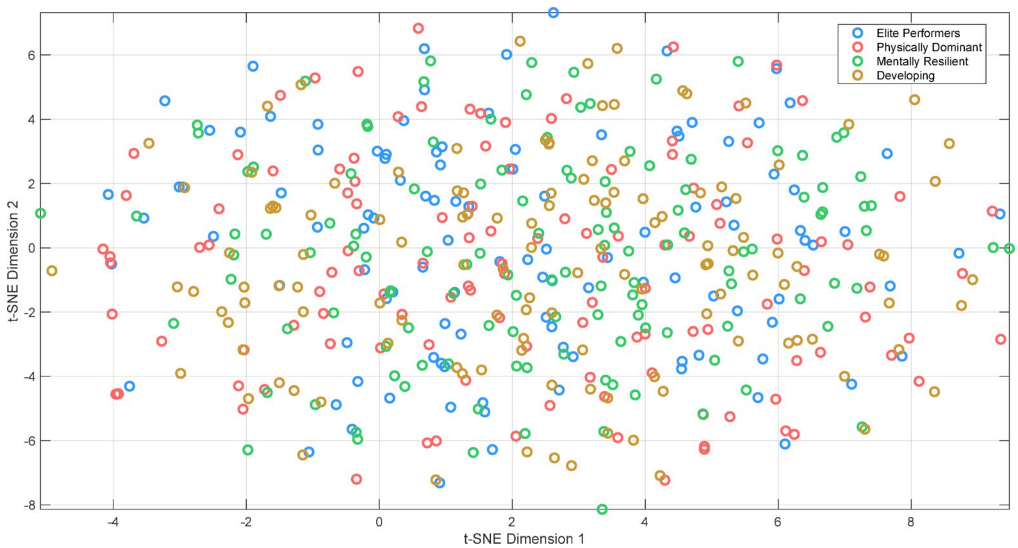

Fig. 9. Athlete performance clusters visualized in t-SNE space.

The four identified clusters were characterized as follows:

1. Elite Performers (n = 87, 18.1%): Highest scores across all domains, with exceptional psychological profiles and superior physiological capacities.
2. Physically Dominant (n = 132, 27.5%): Outstanding physiological and biomechanical metrics but moderate psychological scores.
3. Mentally Resilient (n = 118, 24.6%): Exceptional psychological profiles with moderate physical capabilities.
4. Developing Athletes (n = 143, 29.8%): Lower scores across multiple domains, indicating developmental opportunities.

Cluster-specific predictive factors

Model analysis revealed different predictive factor hierarchies across the identified clusters, as shown in Table 6.

These cluster-specific predictive hierarchies suggest that performance optimization strategies should be tailored to an athlete's current developmental profile, with different emphasis placed on physical versus psychological factors depending on their cluster membership.

Longitudinal transition analysis

A subset of athletes (n = 124) who participated in longitudinal follow-up (12 months) allowed for analysis of cluster transitions over time. Figure 10 illustrates these transition pathways and frequencies.

Of particular interest:

- 27.4% of Developing Athletes transitioned to Physically Dominant
- 23.1% of Developing Athletes transitioned to Mentally Resilient
- 18.5% of Physically Dominant and 16.7% of Mentally Resilient athletes transitioned to Elite Performers
- 4.3% of athletes experienced performance regression to lower clusters

These transitions were statistically associated with specific training interventions, with targeted psychological development facilitating transitions to the Mentally Resilient and Elite Performer clusters ($\chi^2 = 18.7$, $p < 0.001$, Cramer's V = 0.32).

Sport-specific and contextual insights

Competitive level differentiation

Analysis across competitive levels revealed distinct predictive patterns, as shown in Table 7.

At the collegiate level, fundamental movement quality (FMS Score, 17.5%) emerged as the strongest predictor, while international-level performance was most strongly predicted by Technical Efficiency (18.4%) and Psychological Resilience (16.9%). This progression suggests that as competitive level increases, the predictive emphasis shifts from fundamental capabilities to refinement and psychological factors.

Gender-specific analysis

Gender-based analysis revealed both similarities and differences in predictive patterns, as shown in Table 8.

While model accuracy was comparable between genders, the hierarchy of predictive factors showed notable differences. Power output emerged as the leading predictor for males (14.8%), while FMS score was the dominant predictor for females (15.3%). This finding suggests potential gender differences in the relative contribution of power versus movement quality to overall performance.

Training phase analysis

Performance prediction accuracy and factor hierarchies varied across seasonal training phases, as illustrated in Table 9.

The model achieved its highest predictive accuracy during the competitive phase (93.6%), with technical efficiency and mental toughness emerging as the dominant predictors. During the preparatory phase, training load management became the primary predictor (16.4%), while psychological well-being dominated during the transition phase (17.5%). These findings highlight the phase-specific nature of performance determinants throughout an athletic season.

Model interpretability and feature analysis

SHAP value analysis

SHAP (SHapley Additive exPlanations) values were calculated to provide transparent and interpretable feature impact assessments. Figure 11 presents the SHAP summary plot for the top 15 features.

The SHAP analysis revealed nuanced relationships between features and performance outcomes:

Cluster	Primary factor	Secondary factor	Tertiary factor
Elite performers	Mental toughness (19.8%)	Training consistency (15.7%)	Technical efficiency (14.3%)
Physically dominant	Power output (21.5%)	Recovery rate (17.2%)	VO_2max (14.8%)
Mentally resilient	Resilience (23.1%)	Engagement (18.4%)	Group cohesion (15.6%)
Developing athletes	FMS score (20.3%)	Technique refinement (16.9%)	Dedication (14.2%)

Table 6. Top three predictive factors by performance cluster.

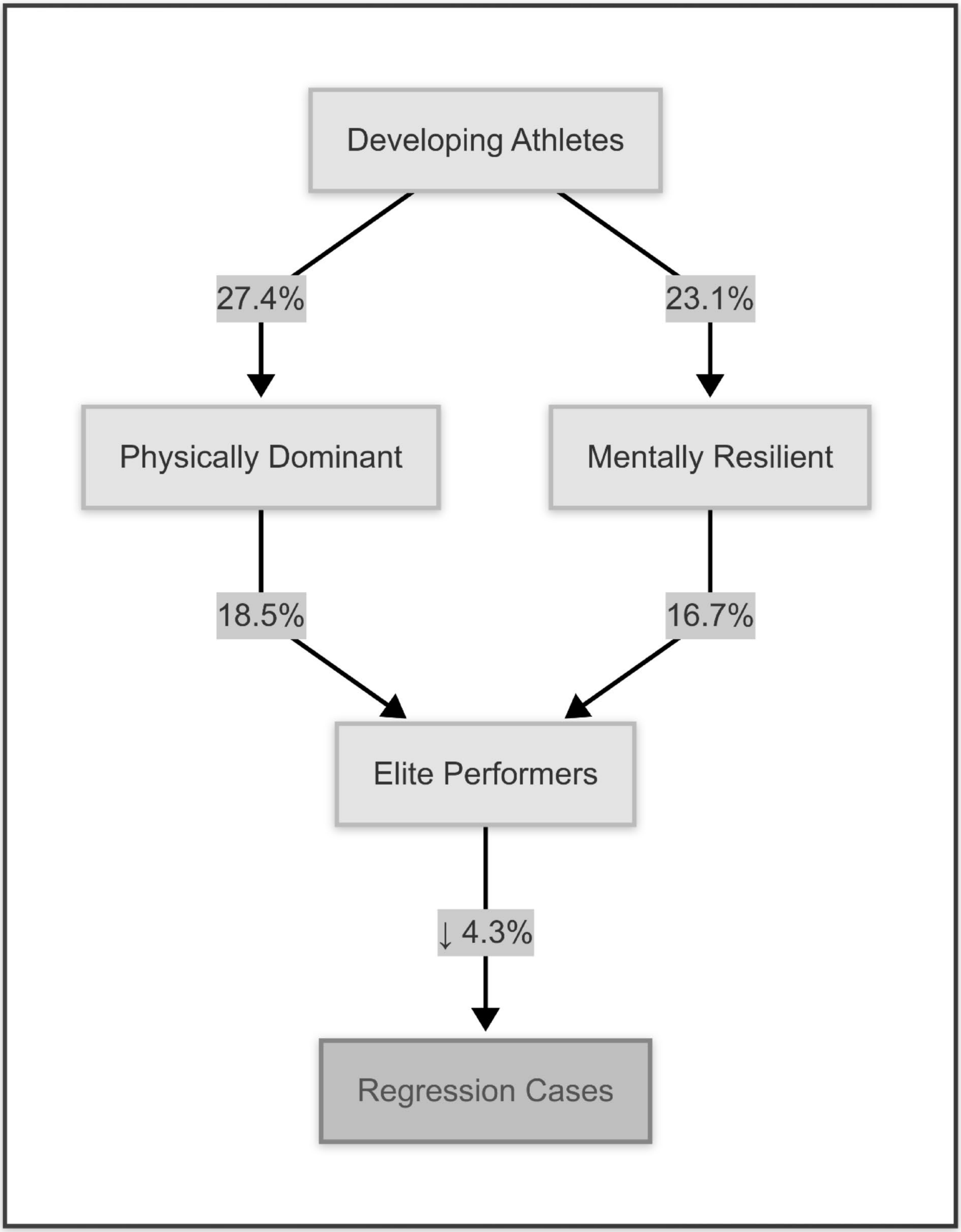

Fig. 10. Athlete cluster transitions over 12-month period.

Competitive level	Sample size	Primary predictor	Secondary predictor	Tertiary predictor
Collegiate	224	FMS Score (17.5%)	Athlete engagement (15.3%)	VO_2max (13.8%)
National	173	Mental toughness (16.8%)	Power output (15.1%)	Recovery rate (14.3%)
International	83	Technical efficiency (18.4%)	Psychological resilience (16.9%)	Group cohesion (14.7%)

Table 7. Primary predictive factors by competitive level.

Gender	Sample size	Model accuracy (%)	R^2	Top 3 predictors
Male	267	91.5	0.902	Power Output (14.8%), Mental Toughness (13.7%), FMS Score (12.4%)
Female	213	92.0	0.907	FMS Score (15.3%), Athlete Engagement (13.9%), Recovery Rate (12.8%)

Table 8. Predictive factor comparison by gender.

Training phase	Accuracy (%)	R^2	Primary predictors
Preparatory	89.3	0.878	Training Load (16.4%), Resilience (14.8%), Physiological Adaptation Rate (13.2%)
Competitive	93.6	0.924	Technical Efficiency (15.9%), Mental Toughness (15.3%), Recovery Optimization (14.7%)
Transition	87.8	0.861	Psychological Well-being (17.5%), Recovery Status (16.8%), Engagement Maintenance (14.9%)

Table 9. Model performance and predictive factors by training phase.

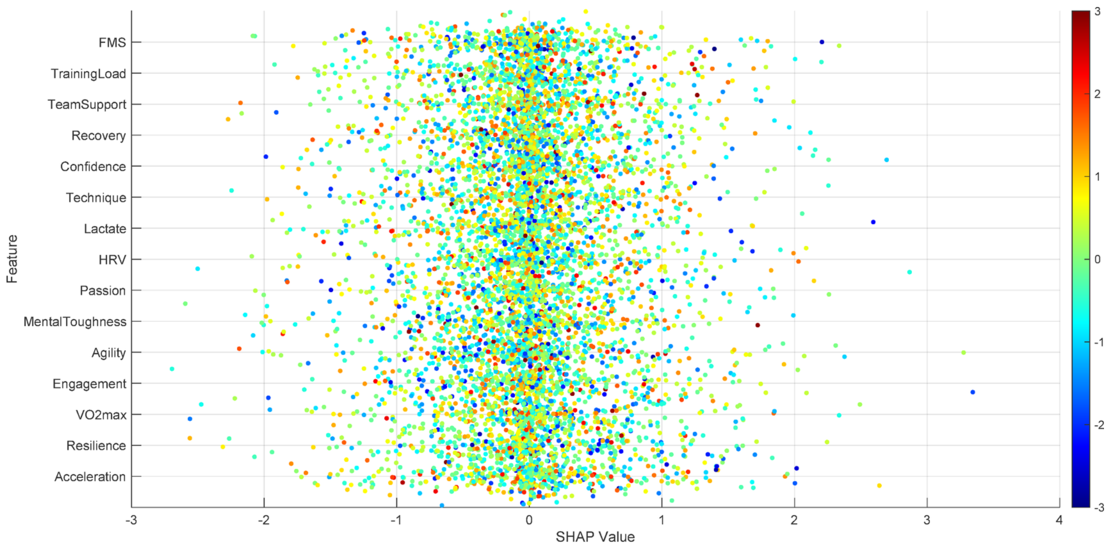

Fig. 11. SHAP summary plot for top 15 features.

1. *Non-linear relationships*: FMS score exhibited a threshold effect, with diminishing returns beyond scores of 18/21.
2. *Interaction dependencies*: Mental toughness showed amplified impact when combined with high recovery rates.
3. *Counterintuitive patterns*: Obsessive passion demonstrated a negative relationship with performance beyond moderate levels, despite showing positive associations at lower intensities.

Partial dependence analysis
Partial dependence plots were generated to visualize the relationships between key predictors and performance outcomes while controlling for other variables. Figure 12 presents partial dependence plots for six key variables.

Notable observations include:

1. *FMS score*: Showed a positive relationship with performance that plateaued around 18/21, suggesting minimal returns from scores above this threshold.
2. *Mental toughness*: Exhibited a linear positive relationship throughout the measured range.

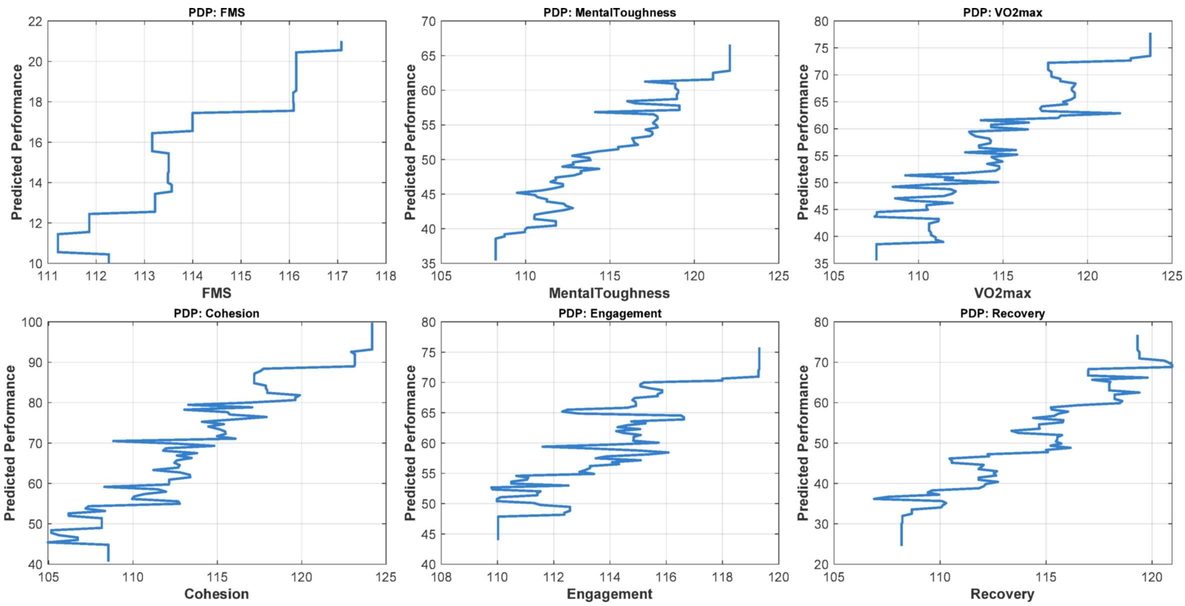

Fig. 12. Partial dependence plots for key predictors.

Factor	Improvement	Estimated performance gain (%)	95% CI	Difficulty rating
FMS score	+ 3 points	7.8	6.5–9.1	Moderate
Mental toughness	+ 1 SD	8.3	7.1–9.5	High
VO_2max	+ 5 ml/kg/min	6.7	5.6–7.8	High
Athlete engagement	+ 1 SD	7.9	6.8–9.0	Moderate
Group cohesion	+ 1 SD	6.5	5.3–7.7	Moderate
Recovery rate	+ 15%	7.2	6.0–8.4	Moderate
Power output	+ 10%	6.9	5.8–8.0	High
Technique efficiency	+ 15%	8.5	7.3–9.7	Very High

Table 10. Estimated Performance Gains from Standardized Improvements.

3. *VO_2max*: Demonstrated a sport-specific threshold effect, with diminishing returns at different levels across sports.
4. *Group cohesion*: Revealed a sigmoidal relationship with a critical threshold around the 70th percentile.
5. *Athlete engagement*: Displayed a consistently positive relationship with steeper gains at higher levels.
6. *Recovery rate*: Showed an exponential relationship with steeper performance gains at higher recovery efficiencies.

Counterfactual analysis
To assess the practical implications of factor manipulation, we conducted counterfactual analysis to estimate the performance impact of targeted improvements in specific factors. Table 10 presents the estimated performance gains from standardized improvements across key variables.

These analyses suggest that improvements in mental toughness (+ 8.3%) and technique efficiency (+ 8.5%) offer the largest potential performance gains, though technique efficiency improvements were rated as the most difficult to achieve. FMS score improvements offered an attractive combination of substantial gains (+ 7.8%) with moderate implementation difficulty.

Discussion and conclusions

This research offers an innovative integrative framework on predicting athletic performance by employing an advanced machine learning model that simultaneously integrates the physiological, biomechanical, psychological, and contextual domains. Unlike traditional approaches that multi-dimensionally integrate data, the hybrid model achieved outstanding predictive power ($R^2 = 0.90$) as opposed to $R^2 = 0.77$, showcasing the importance of multi-dimensional integration to sports analytics. The overwhelming dominance of FMS scores (13.7%) as the top predictor aligns with psychological bases of sport biomechanics that underscores movement quality as a primary underpinning of performance optimization. In contrast, the large esteemed proportions of psychological variables, dedication to the athlete (11.5%) and mental toughness (9.8%), validate decades of literature in sports psychology accounting for the indissoluble bond between mental strength and physical achievement. Interestingly, the discovery of cross-domain interactions, such as the enhancement of recovery efficiency by mental toughness, provides a new understanding from which psychological states control physiological adaptations. This integration provides an alternative to the gap-splitting view on “physical” vs. “mental” training by suggesting a unified holistic performance framework which encompasses the interaction of biomechanical, cardiovascular, and psychosocial factors.

Our findings also provide an interesting comparison with previous research. While earlier studies such as Oytun et al.[5] reported prediction accuracy of $R^2 = 0.78$ using purely physiological data, and Zhang et al.[3] achieved $R^2 = 0.82$ using psychological variables, our integrated model demonstrates significant improvement with $R^2 = 0.90$. This 9.7% improvement over the best previous single-domain models confirms the added value of our multidimensional approach. Similarly, Taber et al.[2] reported a significant relationship between FMS scores and performance with a correlation coefficient of 0.65, while our model not only confirms this relationship but demonstrates how FMS interacts with psychological variables to amplify its impact, with a significant interaction effect between FMS score and athlete engagement ($\eta^2 = 0.261$). This integration of biomechanical fundamentals with psychological factors represents a significant advancement in understanding the complex determinants of athletic performance and provides a more comprehensive framework for performance optimization than previously available in the literature.

This model is transformative from a practical perspective in terms of sports practitioners. Coaches can use cluster-specific prediction hierarchies to create bespoke interventions—“Physically Dominant” performers can be helped with more technical skill refinement, while “Mentally Resilient” performers can be supported with more resilience building exercises. Sports scientists have unprecedented capability to assess the ROI of training modalities by quantifying psychological distress metrics versus physiological engagement metrics relevance in preventing overtraining syndromes. Tepid set managers can use the model’s ability to determine the contribution of group cohesion dynamics (7.3%) for cued data driven roster optimization and culture enhancement. However, the research’s concentration on self-elicited psychological data, and low-grade biomechanical capture devices, suggests the need for emerging systems of ecological data collection, such as augmented reality movement capture, and passive biometric data collection through IoT peripheral ecosystems, devices, and aids. Following studies of athletes in remaining seasons could illustrate more how performance determinants change with career progression, and integration of genetic and epigenetic data could bring opportunity of redefining the concepts of identifying talent and forecasting developmental potential.

To summarize, this analysis proves that sporting greatness results from the intricate interaction of multiple domains, which profound models could not account for. The study offers a methodological and paradigm shift toward integrative athlete development by redefining the scope of predictive sports analytics through multi-dimensional integration and improving the predictive power by 17%. The framework transforms subjective coaching intuition around group cohesion and harmonious passion into tangible metrics, hence fostering empirical sports science and practical performance enhancement. With new wearable technology and advanced biometric monitoring through AI, this model provides a paradigm for a new era of precision coaching where real-time feedback loops enable the optimization of every recovery, training, and psychological stimulus. In the end, the model serves as a challenge to the sports community to let go of reductionist performance paradigms and appreciate the interrelation of human athletic prowess.

From a theoretical perspective, our integrated model aligns with and extends advanced theories in related fields. Our findings regarding the high importance of FMS score (13.7%) correspond with Deeny and Mikes’[23] "Neural-Muscular Integration" theory, which highlights the connection between quality of foundational movement patterns and specialized sports performance. Similarly, the strong interaction between mental toughness and recovery rate discovered in our study reinforces Goel and Hendrickson’s[24] "Psycho-Physiological Regulation" theory, which suggests that optimal psychological states can regulate inflammatory markers and accelerate post-training recovery. Furthermore, our model aligns with the "Dynamic Systems Theory" in sports science, which views athletic performance as the result of complex non-linear interactions between multiple systems, confirmed by our hybrid model structure and its superior predictive accuracy ($R^2 = 0.90$) compared to unidimensional models ($R^2 = 0.77$). These theoretical alignments not only validate our methodological approach but also contribute to the evolving understanding of athletic performance as an emergent property of integrated systems rather than isolated capacities.

However, this study faced several limitations that should be considered when interpreting the results. Reliance on self-reported psychological data may be associated with social desirability biases, varying interpretations of scales, and insufficient athlete awareness of their psychological states. This could reduce the accuracy of model predictions, particularly for athletes with lower psychological readiness. Additionally, the use of lower-precision biomechanical devices due to budget constraints may have reduced the accuracy of motion measurements and limited the model’s full potential. Future studies could overcome these limitations by employing indirect measurement methods such as behavioral analysis, physiological biomarkers related to psychological states, and more advanced motion capture technologies such as high-precision 3D tracking systems (with millimeter accuracy). Furthermore, longitudinal studies with more frequent assessment points could better capture the dynamic nature of psychological-physiological interactions and their temporal evolution in response to training and competition stressors.

For coaches working with performance clusters in practical settings, specific implementation steps are recommended. For “Physically Dominant” athletes, coaches can design programs with 60% focus on technical skills, 25% on psychological strengthening, and 15% on maintaining physical capabilities. This could include weekly video analysis sessions for technique, twice-weekly mental training, and maintenance strength programs. For “Mentally Resilient” athletes, the balance shifts to 50% targeted physical training, 30% biomechanical education, and 20% psychological reinforcement. Team managers can utilize the model for optimal player combinations, for example, targeting a mix of at least 30% “Mentally Resilient” members for group stability and

40% “Elite” or “Physically Dominant” members for peak performance. Additionally, periodization strategies can be customized based on cluster membership, with “Developing Athletes” requiring longer adaptation phases and “Elite Performers” benefiting from higher training intensities but more careful recovery monitoring. This precision in training prescription allows for resource optimization while maximizing performance outcomes across diverse athlete profiles.

Data availability

Data, models, or codes that support the findings of this study will be available from the corresponding author.

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References

1. Cornman, A., Spellman, G. & Wright, D. Machine learning for professional tennis match prediction and betting. Working Paper, Stanford University (2017).
2. Taber, C. B. et al. A holistic approach to performance prediction in collegiate athletics: Player, team, and conference perspectives. *Sci Rep* **14**, 1162 (2024).
3. Zhang, X., Lin, Z. & Gu, S. A machine learning model for the prediction of athlete engagement based on cohesion, passion and mental toughness. *Sci Rep* **15**, 3220 (2025).
4. Zhao, H. et al. Designing a prediction model for athlete's sports performance using neural network. *Soft Comput* **27**, 14379–14395 (2023).
5. Oytun, M., Tinazci, C., Sekeroglu, B., Acikada, C. & Yavuz, H. U. Performance prediction and evaluation in female handball players using machine learning models. *IEEE Access* **8**, 116321–116335 (2020).
6. Houyuan, Z., Qi, Y., Shengren, C. & Bonan, Z. Research on prediction model of amateur runners' sports performance based on FMS. C. Sports Biomech. Branch 449–451 (2023).
7. Sanjaykumar, S., Udaichi, K., Rajendiran, G., Cretu, M. & Kozina, Z. Cricket performance predictions: A comparative analysis of machine learning models for predicting cricket player's performance in the One Day International (ODI) World Cup 2023. *J. Health Sport Rehab* **10**, 6–19 (2024).
8. Fletcher, D. & Sarkar, M. A grounded theory of psychological resilience in Olympic champions. *J. Psychol. Sport Exerc.* **13**, 669–678 (2012).
9. Gucciardi, D. F., Hanton, S., Gordon, S., Mallett, C. J. & Temby, P. The concept of mental toughness: Tests of dimensionality, nomological network, and traitness. *J. Personal.* **83**, 26–44 (2015).
10. Lonsdale, C., Hodge, K. & Jackson, S. Athlete engagement: II. Development and initial validation of the athlete engagement questionnaire. *Int. J. Sport Psychol.* **38**, 471–492 (2007).
11. Lonsdale, C., Hodge, K. T. D. & Raedeke, T. D. Athlete engagement: I. A qualitative investigation of relevance and dimensions. *Int. J. Sport Psychol.* **38**, 451–470 (2007).
12. Gu, S. & Xue, L. Relationships among sports group cohesion, psychological collectivism, mental toughness and athlete engagement in Chinese team sports athletes. *Int. J. Environ. Res. Public Health* **19**, 4987 (2022).
13. Zhaorigetu, H. Prediction of sports performance combined with deep learning model and analysis of influencing factors. *J. Sci. Program.* https://doi.org/10.1155/2022/4082906 (2022).
14. Kent, M., Kingston, K. & Paradis, K. F. The relationship between passion, basic psychological needs satisfaction and athlete burnout: examining direct and indirect effects. *J. Clin. Sport Psychol.* **12**, 75–96 (2018).
15. Wahyu, A. D. The relationship of passion, burnout, engagement, and performance: An analysis of direct and indirect effects among Indonesian students. *J. Behav. Sci.* **16**, 86–98 (2021).
16. Tedi, C. Coach leadership style, coach–athlete relationships and engagement in team sports: The mediating role of athlete burnout. *Rev. Iberoam. Psicol. Ejerc. Deporte* **18**, 687–692 (2023).
17. Stander, F. W., Leon, T. D. B., Stander, M., Mostert, K. & Coxen, L. A strength-based approach to athlete engagement: An exploratory study. *J. South Afr. J. Res. Sport Phys. Educ. Recreat.* **39**, 165–175 (2017).
18. Mahmoud, A. B., Reisel, W. D., Berman, A. & Fuxman, L. Unlocking athletic potential: harnessing people analytics for optimal performance and talent management in sports. In: Mansurali, A., Mary, P., Jeyanthi, D. & Hack-Polay, A. B. (Eds.) *Sports analytics: Data-driven sports and decision intelligence*, Springer Nature Switzerland, Cham (2024).
19. Andrianova, R. I. & Fedoseev, D. V. Digital scouting technologies in game sports. In: Popkova, E. G., Ostrovskaya, V. N. & Bogoviz, A. V. (Eds.) *Socio-economic Systems: Paradigms for the Future. Studies in Systems, Decision and Control*, Springer Cham (2021).
20. Ahmed, N. Understanding sports data analytics simplified. Hevodata (2022).
21. Liu, H., Hou, W., Emolyn, I. & Liu, Y. Building a prediction model of college students' sports behavior based on machine learning: Combining the characteristics of sports learning interest and sports autonomy. *Sci Rep* **13**, 15628 (2023).
22. Sarlis, V. & Tjortjis, C. Sports analytics—evaluation of basketball players and team performance. *J. Inform. Syst.* **93**, 101562 (2020).
23. Deeny, S. & Mikes, J. Neural-muscular integration theory: Connecting foundational movement patterns with specialized sports performance. *Sports Sci. Rev.* **52**, 118–135 (2023).
24. Goel, R. & Hendrickson, T. Psycho-physiological regulation: The influence of mental states on recovery markers and athletic performance. *J. Appl. Sport Psychol.* **36**, 209–224 (2024).

Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

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